Nearsightedness at strong coupling

Exact self-energies versus GW and Kohn–Sham potentials in one-dimensional Wigner molecules and stretched bonds

Research proposal and work plan for a theoretical model-systems paper built on the iDEA code

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Abstract

Kohn's principle of nearsightedness says that local electronic properties depend only weakly on distant changes in the external potential. It underlies linear-scaling methods, embedding theories and every local or semilocal approximation in electronic-structure theory. In earlier work with the iDEA code [1], we showed that the self-energy Σ of many-body perturbation theory (MBPT) is advantageously nearsighted, whereas the exact Kohn–Sham (KS) potential is not. That study used *approximate* self-energies, spinless electrons and weakly correlated systems.

The decisive test is strong coupling: stretched bonds, Mott insulators and Wigner molecules. This is where both density-functional theory (DFT) and GW fail, and where few-electron Wigner physics is now imaged in nanotubes, quantum dots and moiré artificial atoms [2–5].

We propose to compute, for the first time, *exact* one-body Green's functions and self-energies of interacting electrons in the one-dimensional continuum across the crossover into the Wigner regime, and to compare them with GW and with exact KS potentials.

Central hypothesis: whether the exact self-energy is nearsighted is controlled by near-degeneracy and spin.

- It remains nearsighted for non-degenerate subsystems and for same-spin electrons.
- In the strongly correlated singlet regime it develops a low-frequency component that couples distant subsystems, with separation-independent weight: the real-space fingerprint of the Mott pole.
- GW remains nearsighted throughout and misses it.

A closed-form result for the Hubbard dimer, verified numerically, already shows this structure. The project uses iDEA's GPU-accelerated exact solver, the space-time GW implementation in open pull request #3, and a new exact-Green's-function module. It targets *Phys. Rev. B*, or *Phys. Rev. Lett.* if the breakdown proves clean and general. Planned duration is about six months part-time, with under 50 GPU-hours.

1 Background and motivation

1.1 Nearsightedness, and what we showed in 2019

Kohn introduced nearsightedness to justify linear-scaling DFT [6]. Prodan and Kohn sharpened it: at fixed chemical potential, the density at a point responds only to nearby changes in the effective potential [7].

The exact KS potential is *not* nearsighted in this sense:

- For weakly coupled subsystems with different local ionisation energies, the exact exchange–correlation (xc) potential forms spatial steps whose height is set globally [8–10].
- In the symmetric stretched singlet, it forms a peak at the bond midpoint [11, 12].

In Ref. [1], two electrons were confined to separate one-dimensional (1D) wells. The exact KS potential showed a step, whereas Hartree–Fock and GW self-energies did not. When distant subsystems were combined, the self-energy demonstrated nearsightedness numerically. Related nonlocal-potential schemes point the same way: Koopmans-compliant hybrids [13] and generalised KS with a local correlation potential [14].

1.2 Why strong coupling is the decisive test

The 2019 conclusion is only as strong as GW's physics, and GW fails qualitatively in strongly correlated regimes:

- static correlation and delocalisation errors in H_2 dissociation [15, 16];
- the missing vertex in the atomic (Mott) limit of the Hubbard dimer [17];
- self-consistency making results worse for two electrons on a sphere [18].

In this regime the exact self-energy is expected to contain a Mott pole near the chemical potential, exactly the kind of feature that might *not* be nearsighted. The answer matters for:

- local and embedding self-energy schemes, such as dynamical mean-field theory with its local self-energy [19];
- connector theory, which transfers spectral information from models to materials [20–23];
- practical "nearsighted" MBPT-inspired functionals.

Strongly correlated few-electron systems are also experimentally topical [2–5], and are the subject of recent *ab initio* 1D studies [24, 25].

1.3 What is missing

- **Lattice only.** Exact self-energies exist only for lattice models: the Hubbard dimer [17], its spectral potential [23], and exact GW and vertex benchmarks in the extended Hubbard model [26]. A recent analysis argues that the inferred cancellation of vertex corrections needs re-examination [27].
- **No real-space map.** Nobody has mapped the real-space nonlocality of an exact self-energy against separation and correlation strength, side by side with GW and the exact KS potential.
- **Local frequency-dependent alternatives untested.** Neither spectral potentials [20, 22] nor dynamical xc fields [28] have been tested against exact continuum references.

1.4 Why now, and why iDEA

iDEA (v1.1.0) solves the many-electron Schrödinger equation exactly on a 1D grid, for any potential, interaction and spin configuration, with a GPU-accelerated solver. It also provides mean-field methods, an experimental KS strictly-correlated-electrons (KS-SCE) module [29–31] and exact KS inversion. Three developments make this project feasible now:

- **Space-time *GW*.** Open PR #3 adds G_0W_0, GW_0 and fully self-consistent GW in the space-time formalism [32, 33]. It uses optimised imaginary-time and imaginary-frequency grids in the spirit of the discrete Lehmann representation [34], and Galitskii–Migdal energies [35].
- **GPU acceleration.** Three-electron Hamiltonians on grids of about 100–150 points are now practical. This is exactly the $(N+1)$ sector needed for electron addition to a two-electron system.
- **Consolidated fixes.** A set of fixes and many-body observables is ready on feature branches (Sec. 6.1).

2 Central hypothesis and preliminary analytic result

2.1 The Hubbard dimer

For the half-filled Hubbard dimer, $H = -t\sum_\sigma (c^\dagger_{1\sigma}c_{2\sigma} + \text{h.c.}) + U\sum_i n_{i\uparrow}n_{i\downarrow}$, the exact self-energy relative to the non-interacting Hamiltonian is diagonal in the bonding/antibonding basis [17]:

$$\Sigma_{b/a}(\omega) = \frac{U}{2} + \frac{U^2/4}{\omega - \mu \mp 3t}, \qquad \mu = \frac{U}{2}. \tag{1}$$

In the site basis, at imaginary frequency $\omega = \mu + i\nu$,

$$\Sigma_{12}(\mu + i\nu) = -\frac{3tU^2/4}{\nu^2 + 9t^2}, \qquad \int_{-\infty}^{\infty} d\nu\, \Sigma_{12}(\mu + i\nu) = -\frac{\pi U^2}{4}, \qquad \Sigma_{12}(\mu) = -\frac{U^2}{12t}. \tag{2}$$

We verified these expressions by exact diagonalisation in Fock space (Appendix A). The pole position, residue and constant are reproduced to ten digits for $0.05 \le t/U \le 0.2$. The weight $-\pi U^2/4$ is reproduced to quadrature accuracy for $0.01 \le t/U \le 0.2$.

2.2 Interpretation

As the sites separate, t decays exponentially and the inter-site self-energy behaves in three ways at once:

- **Finite frequency:** at fixed $\nu \gg t$, $\Sigma_{12} \approx -3tU^2/(4\nu^2) \propto t$, so it is *nearsighted*.
- **Chemical potential:** $\Sigma_{12}(\mu) \propto 1/t$ grows, so it is *ultra-far-sighted*.
- **Integrated weight:** independent of separation.

The nonlocal part does not disappear. It collapses onto a window of width $\sim 3t$ around μ: the Mott pole viewed in real space.

2.3 Two controls that restore nearsightedness

Asymmetry. Site energies $\pm\Delta/2$ suppress the inter-site self-energy, but only gradually (Table 1).

Spin. For two same-spin electrons on two sites, the ground state is a single determinant and Σ reduces to its static Hartree–Fock form. There is no low-frequency nonlocal component.

Table 1: Asymmetric Hubbard dimer ($U = 1$, μ at midgap): inter-site self-energy and its frequency-integrated weight.

t/U	Δ/U	$\Sigma_{12}(\mu)/U$	$\Sigma_{12}(\mu + 0.3iU)/U$	$\int d\nu\, \Sigma_{12}/U^2$
0.05	0	−1.667	−0.333	−0.785
0.05	0.2	−1.111	−0.303	−0.641
0.05	0.5	−0.390	−0.201	−0.379
0.01	0	−8.333	−0.083	−0.785
0.01	0.05	−4.908	−0.082	−0.603
0.01	0.2	−0.685	−0.074	−0.225
0.01	0.5	−0.117	−0.049	−0.093

2.4 What the continuum adds

The dimer cannot tell us four things:

- the real-space range and shape of the nonlocal component;
- the role of long-range Coulomb tails and of the internal structure of Wigner molecules;
- the quantitative comparison with the exact KS potential (steps and peaks);
- whether GW captures any of it.

3 Research questions and hypotheses

RQ1 (exact Σ in the 2019 setting).
Question: in an asymmetric double well with one electron per well, is the exact $\Sigma(x, x'; i\nu)$ nearsighted, for both singlet and triplet?
H1: for asymmetry $\Delta \gg t_{\text{eff}}$, the exact Σ is nearsighted at all frequencies while the exact v_{xc} shows the step; singlet and triplet behave alike.
Falsified by: a sizeable inter-well block at low ν.

RQ2 (symmetric dissociation; central).
Question: how does the inter-subsystem block $\Sigma_{AB}(i\nu)$ scale with separation d and frequency, and does it collapse onto Eq. (2) with $t_{\text{eff}}(d)$ and U_{eff} extracted from the exact spectra?
H2: in the singlet,

- $\|\Sigma_{AB}\|$ at fixed ν decays like $t_{\text{eff}}(d)$;
- $\Sigma_{AB}(\mu)$ grows like $1/t_{\text{eff}}$;
- the integrated weight is d-independent, $\approx \pi U_{\text{eff}}^2/4$ on localised orbitals.

The triplet shows no such component, and asymmetry restores nearsightedness for Δ between $\sim t_{\text{eff}}$ and U_{eff}.
Falsified by: a weight that decays with d, or a triplet with the same low-frequency component.

RQ3 (Wigner molecule in a single soft well).
Question: does the spatial range of Σ track the Wigner spacing as the coupling λ grows, and at which frequencies?
H3 (exploratory): the range grows with λ and concentrates at low ν, driven by near-degeneracies of the $(N+1)$ Wigner molecule. The GW range, set by Hartree–Fock orbitals, does not grow correspondingly.

RQ4 (approximations).
Question: how do restricted and unrestricted Hartree–Fock, G_0W_0@HF, GW_0 and self-consistent GW compare with the exact nonlocality?
H4:

- Spin-restricted GW is spuriously nearsighted: it has no Mott pole, so the gap is too small.
- Unrestricted-HF-based GW opens the gap by static symmetry breaking, which is nearsighted but spin-contaminated. This is the MBPT face of the symmetry dilemma [36].
- The exact v_{xc} is far-sighted in every case. KS-SCE reproduces this at strong coupling [37, 38].

RQ5 (consequences).
Question: what is lost by *enforcing* nearsightedness, and what is the minimal far-sighted ingredient?
H5: truncating Σ to intra-subsystem blocks is harmless for the triplet and asymmetric cases but closes the gap in the symmetric singlet. A rank-2 low-frequency "Mott-pole" coupling, with parameters from local quantities, restores the ionisation potential (IP), electron affinity (EA) and gap.

4 Model systems and parameter space

Interactions. The default is iDEA's softened $v(x, x') = 1/(|x - x'| + 1)$ [1]. The soft-Coulomb $1/\sqrt{(x-x')^2+1}$ is a check. A short-range (Yukawa-screened) matrix is a control that separates genuine correlation nonlocality from the Coulomb tail.

Units. Atomic units are used throughout. In effective-mass units (e.g. GaAs: $\mathrm{Ha}^* \approx 11\,\mathrm{meV}$, $a_B^* \approx 10\,\mathrm{nm}$) the same models describe quantum-dot and nanotube devices.

Grids. $N_x \approx 80$–150 with the 13-point stencil. The parameters of S1 will be matched to Ref. [1] from its published repository (JackWetherell/advantageous-nearsightedness).

System	External potential	N, spin	Scan	Probes
S1 asymmetric double well	two softened-Coulomb or Gaussian wells, $V_A \neq V_B$, separation d	2, `ud`/`uu`	$d = 2$–12; Δ from 0 to $O(U_{\mathrm{eff}})$	RQ1, RQ2, RQ4
S2 symmetric double well	as S1, $V_A = V_B$	2, `ud`/`uu`	d until $t_{\mathrm{eff}} \sim 10^{-5}$	RQ2, RQ4, RQ5
S3 single soft well	$\frac{1}{2}\omega^2 x^2$	2, `ud`/`uu`	$\lambda = \omega^{-1/2} = 1$–$6$	RQ3, RQ4
S4 (stretch goal)	as S1–S3	3, removal sector only	—	three-electron photoemission

5 Methods

5.1 Exact ground states

`iDEA.methods.interacting.solve(s, k, GPU=True)` returns Ψ_0^N, and $\langle S^2 \rangle$ labels the spin states. For $N = 2$ in the `ud` configuration, the spatial wavefunction $\psi(x_1, x_2)$ (electron 1 up, electron 2 down) fully specifies the state.

5.2 Exact one-body Green's function

We use the conventions of PR #3: zero temperature, real imaginary-time functions, energies measured from μ. For spin σ,

$$G_\sigma(x, x'; \tau) = -\theta(\tau) \sum_n f_n(x) f_n^*(x') e^{-(\varepsilon_n^+ - \mu)\tau} + \theta(-\tau) \sum_m g_m(x) g_m^*(x') e^{-(\varepsilon_m^- - \mu)\tau}, \tag{3}$$

where

- $\varepsilon_n^+ = E_n^{N+1} - E_0^N$ and $f_n(x) = \langle \Psi_0^N | \hat{\psi}_\sigma(x) | \Psi_n^{N+1} \rangle$;
- $\varepsilon_m^- = E_0^N - E_m^{N-1}$ and $g_m(x) = \langle \Psi_m^{N-1} | \hat{\psi}_\sigma(x) | \Psi_0^N \rangle$.

Removal sector (exact and complete for $N = 2$). The $(N-1)$ states are the one-body eigenstates ϕ_m. Removing an up electron gives $g_m(x) = \int dx_2\, \phi_m^*(x_2) \psi(x, x_2)$, with $\sum_m |g_m(x)|^2 = n_\uparrow(x)$. This is the same algebra as `mbpt.non_interacting_green_function`.

Addition sector (Krylov). We evaluate

$$G_\sigma^+(x, x'; i\omega) = \langle \chi_x | \, [\, i\omega + \mu - (\hat{H}^{N+1} - E_0^N)\,]^{-1} | \chi_{x'} \rangle, \qquad |\chi_x\rangle = \hat{\psi}_\sigma^\dagger(x) |\Psi_0^N\rangle. \tag{4}$$

- $\hat{H}^{N+1}$ is iDEA's exact three-electron Hamiltonian.
- The source is antisymmetric in the like-spin coordinates, so Lanczos remains in the correct sector.
- One Lanczos run per source, with projections onto all $\langle \chi_x |$, gives G^+ at every imaginary time and frequency on PR #3's `SpaceTimeGrid`, and on the real axis with broadening η.

No analytic continuation needed. The IP and EA follow from the large-$|\tau|$ decay of $G(\tau)$; the density from $G(x, x; 0^-)$; the total energy from the Galitskii–Migdal formula.

5.3 Exact self-energy by Dyson inversion

With exact n and one-body density matrix γ,

$$\Sigma_c^{\text{exact}}(i\omega) = i\omega + \mu - \hat{h}_0 - v_H[n] - \Sigma_x[\gamma] - G^{-1}(i\omega), \tag{5}$$

which vanishes as $\omega \to \infty$, because the static limit of the exact Σ is the Hartree–Fock operator of the exact γ. On the grid this inverts PR #3's `dyson_equation` (kernel $\times\, \delta x$ convention):

$$\Sigma_c = [(i\omega + \mu)\mathbb{1} - H - (G\,\delta x)^{-1}]/\delta x. \tag{6}$$

5.4 Exact Kohn–Sham potentials

- **Two-electron singlet (analytic):** $v_{\text{KS}}(x) = \varepsilon_H + \frac{1}{2}\partial_x^2\sqrt{n}/\sqrt{n}$, with $\varepsilon_H = -I$.
- **Otherwise:** `reverse_engineering.reverse(s, n, non_interacting)`.

In both cases $v_{\text{xc}} = v_{\text{KS}} - v_{\text{ext}} - v_H$.

5.5 Approximations compared

- Hartree–Fock, restricted and unrestricted.
- G_0W_0@HF, GW_0 and self-consistent GW (`mbpt.solve`), from restricted and unrestricted starting points.
- LDA and hybrid, as local and semilocal references.
- KS-SCE, as a far-sighted functional that is exact at strong coupling.

5.6 Nearsightedness diagnostics

Subsystems are defined by smooth partitions $P_A + P_B = 1$ and by localised orbitals $w_{A,B}$ built from natural orbitals.

M1, transferability:
$\|P_A(\Sigma^{AB} - \Sigma^A)P_A\|/\|P_A\Sigma^A P_A\|$, with Σ^A the exact self-energy of subsystem A alone. KS counterpart: $\|P_A(v_{\text{xc}}^{AB} - v_{\text{xc}}^A)\|$ in the physical gauge.

M2, range:
Hilbert–Schmidt norm $\|\Sigma_{AB}(i\nu)\|$ against d and ν, giving a decay length $\xi_\Sigma(\nu)$. KS counterpart: step or peak height against d.

M3, response:
$\delta\Sigma(x, x'; i\nu)/\delta v_{\text{ext}}(y)$ for y in the distant subsystem. KS counterpart: $\delta v_{\text{xc}}(x)/\delta v_{\text{ext}}(y)$, alongside $\delta n(x)/\delta v_{\text{ext}}(y)$.

M4, weight:
$\int d\nu\, \langle w_A|\Sigma(i\nu)|w_B\rangle$, compared with $-\pi U_{\text{eff}}^2/4$.

5.7 Consequence tests (RQ5)

- **Truncation:** $\Sigma^{\text{near}} = P_A\Sigma P_A + P_B\Sigma P_B$, solved in Dyson's equation and compared with the exact n, IP, EA, gap and Galitskii–Migdal energy.
- **Pole correction:** add $\Sigma_{AB}^{\text{pole}}(i\nu) = -\frac{3t_{\text{eff}}U_{\text{eff}}^2/4}{\nu^2+9t_{\text{eff}}^2}(|w_A\rangle\langle w_B| + |w_B\rangle\langle w_A|)$.
- **Parameters:** t_{eff} from the $(N-1)$ bonding–antibonding splitting and U_{eff} from the subsystem IP−EA.

5.8 Validation (automated tests)

(i) Zero interaction gives $\Sigma \equiv 0$.

(ii) Small-grid ($N_x \approx 20$–25) Lehmann references from full diagonalisation of the $N \pm 1$ sectors.

(iii) $G(x, x'; 0^-) = \gamma(x, x')$.

(iv) The jump $G(0^+) - G(0^-) = -\delta(x - x')/\delta x$.

(v) $\Sigma(i\omega \to \infty) \to v_H + \Sigma_x[\gamma]$.

(vi) The Galitskii–Migdal energy equals the exact eigenvalue.

(vii) One-electron checks, linking to self-interaction studies of GW [39, 40].

(viii) The dimer limit (Eq. 2) for deep, separated wells.

6 Implementation in iDEA

6.1 Prerequisites (WP0)

Item	Where	Why needed
Space-time GW, `SpaceTimeGrid`, `dyson_equation`, Galitskii–Migdal energy	PR #3 (`iDEA/methods/mbpt.py`)	GW comparisons; common grids and conventions
Restricted HF and hybrid exchange was doubled	`fix/40-rhf-exchange`	restricted starting points
`antisymmetrize` misaligned `space`/`spin` with `full`	`fix/42-antisymmetrize-alignment`	correct spatial parts of states (used to build sources)
Many-body kinetic and interaction energies	`feat/45-manybody-observables`	energy decompositions; Galitskii–Migdal checks
$\langle S_z \rangle$, $\langle S^2 \rangle$	`feat/46-spin-observables`	spin labels of $N \pm 1$ states
Indexing into `ManyBodyStates`	`feat/51-manybodystates-index`	brute-force Lehmann validation
Hartree and KS-SCE energy fixes	`ccr-6064c011-6imjt7`	correct mean-field and KS-SCE energies
KS-SCE checks (default `interp_vsce`, parity, $1/n$ in tails)	new	KS-SCE comparisons; validate against Ref. [30]

6.2 New code

- **`iDEA/methods/exact_green.py`:** `green_function(s, state, grid, ..., GPU)` returns an object compatible with `QuasiParticleState`, so existing observables and `mbpt.total_energy` work unchanged. Also `self_energy` and `spectral_function`.
- **`iDEA/reverse_engineering.py`:** the analytic two-electron singlet KS potential.
- **Tests and tutorial:** `tests/test_exact_green.py` and `tutorial/example_exact_green.ipynb`.
- **Research scripts:** diagnostics and `make_figures.py` in `paper/scripts/`, following the draft CPC paper on branch `ccr-6064c011-6imjt7`.

6.3 Computational cost

At $N_x = 130$:

- the $N = 2$ ground state (dimension 1.7×10^4) takes seconds;
- the three-electron sector has dimension 2.2×10^6 and about 8×10^7 non-zeros (about 1 GB);

- N_x Lanczos runs of 200–400 steps take minutes on an RTX 4090, or about an hour on 4 CPU cores;
- Dyson inversions are negligible;
- GW takes about a minute per system.

About 300 exact-G runs, including convergence checks, fit in under 50 GPU-hours.

7 Work packages and timeline

WP	Content	Milestone
WP0 Consolidation	merge PR #3 and branches; validate KS-SCE; reproduce Ref. [1] with spin	MS1: 2019 results reproduced
WP1 Exact Green's function	`exact_green.py`, removal and addition sectors, tests (i)–(viii)	MS2: tests pass; Galitskii–Migdal energy exact to $\leq 10^{-8}$ Ha
WP2 Diagnostics	M1–M4; truncation and pole tests	MS3 (go/no-go): collapse onto Eq. (2)? Yes: letter; no: detailed PRB
WP3 Production	S1–S3 scans, GW variants, KS, KS-SCE, convergence	MS4: converged dataset
WP4 Consequences	RQ5; optional exact spectral potential [23]; thesis GW material	MS5
WP5 Paper	manuscript, `paper/`, data, README entry	MS6: submission

	M1	M2	M3	M4	M5	M6
WP0	■					
WP1	■	■				
WP2		■	■			
WP3			■	■		
WP4				■	■	
WP5					■	■

8 Expected results and planned figures

1. **Systems and exact KS potentials:** step in the asymmetric case, midpoint peak in the symmetric singlet, none for the triplet.
2. **Self-energy maps:** $\Sigma_{xc}(x, x'; i\nu)$, exact versus GW versus Hartree–Fock, for S1–S3.
3. **Range versus separation:** $\|\Sigma_{AB}(i\nu)\|$ and the integrated weight against d, for singlet, triplet and asymmetric cases, overlaid with v_{xc} step and peak heights.
4. **Mott-pole collapse:** $\langle w_A|\Sigma(i\nu)|w_B\rangle$ rescaled by t_{eff} and U_{eff} onto $-(3/4)/(\tilde{\nu}^2+9)$, with the asymmetry crossover.
5. **Consequences:** spectral functions and IP, EA and gap from the exact, GW, truncated and pole-corrected Σ.

A positive outcome gives a sharp statement: *nearsightedness of the self-energy is protected by non-degeneracy and spin, and fails in a characteristic low-frequency, separation-independent way in the Mott/Wigner regime.* A negative outcome would strengthen the case of Ref. [1] for MBPT-based local approximations at all correlation strengths.

9 Novelty and positioning

Prior work	Established	New here
Ref. [1]	approximate Σ nearsighted; exact KS step	exact Σ; spin; strong coupling; frequency resolution
Refs. [17, 23]	exact dimer Σ and spectral potential	real-space continuum; separation and coupling scans
Refs. [26, 27]	lattice benchmarks; vertex cancellation questioned	exact continuum Σ against the GW hierarchy
Refs. [15, 16, 18]	GW energy and density failures at strong correlation	nonlocality of exact versus approximate Σ
Ref. [41]	real-space nonlocality of the GW Σ in semiconductors	nonlocality of the exact Σ at strong coupling
Refs. [6, 7]	nearsightedness of the density	nearsightedness of Σ at strong coupling
Refs. [8–12]	steps and peaks of the exact KS potential	side by side with the exact Σ
Refs. [29, 37, 38]	strong-coupling KS reproduces steps and peaks	far-sighted KS-SCE versus the exact Σ

10 Risks and mitigations

- **Dyson-inversion round-off:** G to near machine precision; checks of the Galitskii–Migdal energy, density and static limit; reduced natural-orbital basis.
- $(N+1)$**-sector grid limit** ($N_x \lesssim 150$): compact parameters, 13-point stencil, GPU, block Lanczos.
- **Coulomb tails:** analyse Σ_{xc}; short-range control runs.
- **Self-consistent** $\boldsymbol{GW}$ **non-convergence:** damping, G_0W_0 and GW_0 fallbacks; report as a result.
- **KS-SCE defects:** WP0 validation; otherwise non-self-consistent SCE on exact densities.
- **Scope creep:** extensions only after MS3.

11 Outputs, dissemination and follow-ups

Outputs.

- A PRB paper, or a PRL plus a long PRB.
- An exact-Green's-function module in iDEA with tests and a tutorial.
- A reproducible `paper/` directory with data and a README entry.

Follow-ups.

1. Exact continuum spectral potentials (connector theory [20, 21, 23]) and the dynamical xc field [28].
2. Learnability of nearsighted targets versus the hypersensitivity of density-based learning [42].
3. Nearsighted generalised-KS correlation potentials [14].
4. Three-electron removal spectra and real-time extensions.

Suggested collaborators.

- M. J. P. Hodgson and R. W. Godby (York).
- M. Gatti and L. Reining (LSI, École Polytechnique).
- P. Romaniello and J. A. Berger (Toulouse).
- D. P. Kooi and P. Gori-Giorgi (KS-SCE).

12 Alternatives considered

- **The exchange blind spot:** exact singlet–triplet exchange in 1D Wigner molecules against DFT, KS-SCE and GW, including exact time-dependent spin-KS potentials for an exchange gate. Spin enters the strong-coupling functional only at order $e^{-\sqrt{\lambda}}$ [43].
- **Machine-learned non-adiabatic xc potentials:** a crowded area [44].
- **Other frontier ideas:** real-time KS-SCE against exact time-dependent KS potentials; exact ensemble DFT in the Wigner regime; exact thermal (Mermin) DFT; KS versus many-body localisation tensor.

A Hubbard-dimer verification

We built the 16-dimensional Fock space with Jordan–Wigner operators, took the $N = 2$ ground state, formed the exact $G_{ij}(z)$ from the Lehmann sums over $N \pm 1$, and computed $\Sigma = G_0^{-1} - G^{-1}$.

- **Bonding-channel pole:** fitting $\Sigma_b(\omega) = A + R/(\omega - \mu - p)$ gives $A = U/2$, $p = 3t$ and $R = U^2/4$ to ten digits for $t/U = 0.05$, 0.1 and 0.2.
- **Inter-site element:** $\Sigma_{12}(\mu + i\nu)$ matches Eq. (2) at $\nu = 0$ and $10t$ for $0.01 \leq t/U \leq 0.2$, and is real to $\sim 10^{-15}$.
- **Integrated weight:** $-0.78541\,U^2$ for every t, against $-\pi U^2/4 = -0.78540\,U^2$.

Table 1 uses $\mu = [E_0^{N+1} - E_0^{N-1}]/2$.

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